Practice Set B

Class 12 Mathematics, Chapter 7: Integrals

Maximum marks: **54** **20** questions

Name _____ Date _____ Score _____ / 54

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1

8 marks

1 MCQ, NCERT Exemplar — 1 mark

$\int e^x \left(\frac{1-x}{1+x^2}\right)^2 dx$ is equal to:

(A) $\frac{e^x}{1+x^2}+C$ **(B)** $\frac{-e^x}{1+x^2}+C$

(C) $\frac{e^x}{(1+x^2)^2}+C$ **(D)** $\frac{-e^x}{(1+x^2)^2}+C$

2 MCQ, NCERT Exemplar — 1 mark

If $\int \frac{x^3}{\sqrt{1+x^2}}\,dx = a\left(1+x^2\right)^{3/2} + b\sqrt{1+x^2} + C$, then:

(A) $a=\frac{1}{3},\ b=1$ **(B)** $a=-\frac{1}{3},\ b=1$

(C) $a=-\frac{1}{3},\ b=-1$ **(D)** $a=\frac{1}{3},\ b=-1$

3 MCQ, NCERT Misc. Ex. Q37 — 1 mark

The value of $\int_0^1 \sin^{-1} x\,dx$ is:

(A) $\frac{\pi}{2}$ **(B)** $\frac{\pi}{2}-1$ **(C)** $\frac{\pi}{2}+1$ **(D)** 1

4 MCQ, PYQ-style — 1 mark

For $x>0,\ a>0,\ a\neq 1$, $\int \left(e^{x\log a}+e^{a\log x}+e^{a\log a}\right)dx$ is equal to:

(A) $a^x\log a+\frac{x^{a+1}}{a+1}+a^a x+C$ **(B)** $\frac{a^x}{\log a}+\frac{x^{a+1}}{a+1}+a^a x+C$

(C) $\frac{a^x}{\log a}+\frac{x^{a+1}}{a+1}+\frac{a^{a+1}}{a+1}+C$ **(D)** $\frac{a^x}{\log a}+ax^{a-1}+a^a x+C$

5 MCQ, Board-pattern — 1 mark

If $\int_0^{\pi/4} \tan^4 x\,dx = \alpha$, then $\int_0^{\pi/4} \tan^6 x\,dx$ is equal to:

(A) $\alpha-\frac{1}{5}$ **(B)** $\frac{1}{5}-\alpha$ **(C)** $\frac{1}{5}+\alpha$ **(D)** $\frac{1}{6}-\alpha$

Q No.	Question	Marks
6	Assertion-Reason, Board-pattern **Assertion (A):** $\int_{-\pi/2}^{\pi/2} \left(x^3 + x\cos x + \sin^5 x + 1\right) dx = \pi$. **Reason (R):** For every function f continuous on $[-a, a]$, $\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). **(C)** Assertion (A) is true, but Reason (R) is false. **(D)** Assertion (A) is false, but Reason (R) is true.	1 mark
7	Short answer, NCERT Misc. Ex. Q7 Find: $\int \frac{\sin x}{\sin(x-a)}\,dx$.	2 marks
Level 2		**19 marks**
8	Short answer, Board-pattern Evaluate: $\int_{0}^{\pi/2} \cos x \cos 2x \cos 3x\,dx$.	2 marks
9	Short answer, Board-pattern Given $\frac{d}{dx}F(x) = \frac{x}{\sqrt{1+2x^2-x^4}}$ and $F(1) = 0$, find $F(x)$.	2 marks
10	Short answer, NCERT Ex 7.5 Q12 Find: $\int \frac{x^3 + x + 1}{x^2 - 1}\,dx$.	3 marks
11	Short answer, Board-pattern Evaluate: $\int_{-\pi/2}^{\pi/2} \frac{\cos x}{1+e^x}\,dx$.	3 marks
12	Short answer, NCERT Misc. Ex. Q24 Evaluate: $\int_{\pi/2}^{\pi} e^x \left(\frac{1-\sin x}{1-\cos x}\right) dx$.	3 marks
13	Short answer, CBSE SQP 2021-22 (Term 2) Evaluate: $\int_{-1}^{2} \left\lvert x^3 - 3x^2 + 2x \right\rvert dx$.	3 marks
14	Short answer, NCERT Exemplar Find: $\int \frac{\sin^6 x + \cos^6 x}{\sin^2 x \cos^2 x}\,dx$.	3 marks

Q No.	Question	Marks
Level 3		**27 marks**
15	Long answer, NCERT Example 42 Evaluate: $\int_0^{\pi} \frac{x}{a^2\cos^2 x + b^2\sin^2 x}\,dx$, where $a > 0, b > 0$.	5 marks
16	Long answer, NCERT Misc. Ex. Q6 Find: $\int \frac{5x}{(x+1)(x^2+9)}\,dx$.	5 marks
17	Long answer, NCERT Ex 7.10 Q16 Evaluate: $\int_0^{\pi} \log(1+\cos x)\,dx$.	5 marks
18	Short answer, NCERT Exemplar Find: $\int \sqrt{\frac{a+x}{a-x}}\,dx$, where $a > 0$ and $-a < x < a$.	3 marks
19	Long answer, NCERT Exemplar Find: $\int \sqrt{\tan x}\,dx$.	5 marks
20	Case study, Board-pattern **Case Study:** In a mathematics quiz, a team is told to use only the property $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$ (with standard integrals) to evaluate the integrals below. Answer the following. **(i)** Evaluate $\int_0^{\pi/2} \log\left(\frac{4+3\sin x}{4+3\cos x}\right)dx$. — 1 mark **(ii)** Evaluate $\int_0^2 x\sqrt{2-x}\,dx$. — 1 mark **(iii)** Evaluate $\int_0^{\pi/2} \frac{\sin^2 x}{\sin x + \cos x}\,dx$. — 2 marks	4 marks

Answer key

Chapter 7: Integrals, Practice Set B. Full step-wise solutions are in the solutions sheet.

Level 1

1	(A) $\dfrac{e^x}{1+x^2}+C$
2	(D) $a=\dfrac{1}{3},\ b=-1$
3	(B) $\dfrac{\pi}{2}-1$
4	(B) $\dfrac{a^x}{\log a}+\dfrac{x^{a+1}}{a+1}+a^a x+C$
5	(B) $\dfrac{1}{5}-\alpha$
6	(C) A is true, but R is false
7	$x\cos a+\sin a\,\log\lvert\sin(x-a)\rvert+C$

Level 2

8	$\dfrac{\pi}{8}$
9	$F(x)=\dfrac{1}{2}\sin^{-1}\left(\dfrac{x^2-1}{\sqrt{2}}\right)$
10	$\dfrac{x^2}{2}+\dfrac{3}{2}\log\lvert x-1\rvert+\dfrac{1}{2}\log\lvert x+1\rvert+C$
11	1
12	$e^{\pi/2}$
13	$\dfrac{11}{4}$
14	$\tan x-\cot x-3x+C$

Level 3

15	$\dfrac{\pi^2}{2ab}$
16	$-\dfrac{1}{2}\log\lvert x+1\rvert+\dfrac{1}{4}\log\left(x^2+9\right)+\dfrac{3}{2}\tan^{-1}\dfrac{x}{3}+C$
17	$-\pi\log 2$
18	$a\sin^{-1}\dfrac{x}{a}-\sqrt{a^2-x^2}+C$
19	$\dfrac{1}{\sqrt{2}}\tan^{-1}\left(\dfrac{\tan x-1}{\sqrt{2\tan x}}\right)+\dfrac{1}{2\sqrt{2}}\log\left\lvert\dfrac{\tan x-\sqrt{2\tan x}+1}{\tan x+\sqrt{2\tan x}+1}\right\rvert+C$
20	(i) 0 (ii) $\dfrac{16\sqrt{2}}{15}$ (iii) $\dfrac{1}{\sqrt{2}}\log\left(\sqrt{2}+1\right)$